



Spatial and temporal patterns of Landsat-based detection of tree mortality caused by a mountain pine beetle outbreak in Colorado, USA



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ABSTRACT

Mountain pine beetles (*Dendroctonus ponderosae* Hopkins) have killed billions of trees in the United States and Canada in the last decades, thereby causing major alterations to forest ecosystems. Therefore, monitoring the extent and patterns of these major disturbance events are important for subsequent forest management. At 30-m spatial resolution, Landsat imagery affords the opportunity for quantifying spatial and temporal patterns of bark beetle outbreaks at plot to stand resolution. We developed a continuous measure of bark beetle-caused tree mortality using multi-temporal Landsat data (1996–2011) in the Rocky Mountains of northcentral Colorado. We report the year of detection of tree mortality, which is 1 year after attack by beetles. Two approaches were used to predict percent red stage tree mortality within a 30-m grid cell, multiple linear regression models and generalized additive models (GAM) with a nonlinear spatial term. Both models explained >75% of the variance using three Landsat spectral explanatory variables. We used the linear model to predict tree mortality across the entire study area and time series because of its simplicity, its capability for extrapolation beyond the training area, and similar performance compared with the GAM. From 1996 to 2011, cumulative tree mortality was 22% of forested areas within the Landsat scene, equivalent to 228 million mature lodgepole pine trees (range: 174–332 million trees using the 95% confidence interval of field-measured crown areas). Early in the outbreak, tree mortality was associated with expansion, whereas later in the outbreak, mortality was due to intensification (increase in mortality within areas already having beetle activity). We used three metrics (cumulative tree mortality, duration of tree mortality, and average rate of tree mortality) to investigate the temporal and spatial patterns of bark beetle-caused mortality using different grid cell resolutions within the Landsat scene. At the Landsat spatial resolution (900 m²), we estimated that grid cells within the outbreak experienced means of 60% mortality, a duration of 3–4 years, and an average rate of mortality of 20%/year. With coarser spatial resolution, cumulative tree mortality and rate of tree mortality decreased whereas duration of tree mortality increased. Our results improve the understanding of spatial and temporal patterns of tree mortality caused by bark beetle outbreaks, and provide specific information for forest managers and scientists about a severe mountain pine beetle infestation in northcentral Colorado.

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1. Introduction

The mountain pine beetle (*Dendroctonus ponderosae* Hopkins) is one of the most destructive bark beetles in western North America and has killed billions of trees in the United States and Canada in the last few decades (Meddens et al., 2012). Bark beetle-caused tree mortality modifies forest successional patterns (Sibold et al., 2007), alters forest fuel arrangements (Jenkins et al., 2008; Simard et al., 2011; Hicke et al., 2012), changes water yield and peak

snowpack levels (Biederman et al., 2012; Pugh and Gordon, 2012; Pugh and Small, 2012), and reduces carbon uptake by decreasing primary production of live trees (Kurz et al., 2008; Pfeifer et al., 2011) and increasing heterotrophic respiration (Edburg et al., 2011; Moore et al., 2013). Therefore, documenting spatial and temporal patterns of insect disturbances are important for understanding ecosystem impacts and formulating management strategies, e.g., mitigating tree hazards, decreasing immediate crown fire hazard, and utilizing materials from beetle-killed forests (USDA Forest Service, 2011).

Approaches to study landscape-scale patterns of beetle-caused tree mortality include aerial survey by observers in plane (USDA Forest Service, 2012), oblique aerial photography (Macfarlane

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et al., 2013), and digital remote sensing (Wulder et al., 2006a). For bark beetle outbreaks, these methods rely on a typical progression of trees following attack. In the case of mountain pine beetles, water transport in the host tree shuts down rapidly, leading to reduced carbon uptake and eventually tree death (Edburg et al., 2012). The following spring/summer the moisture content of the needles of the killed trees are reduced and the needles start to turn red (red stage). In the years following the start of the red phase, the needles drop to the ground, and when no needles remain (3–5 years) the trees are in the gray stage (Wulder et al., 2006a). In the subsequent decades the killed trees fall to the ground and the forest recovers, leading to increased productivity as well as heterotrophic respiration (Edburg et al., 2011).

Remote sensing images from satellite and aerial platforms have successfully been employed to assess extent and patterns of insect-caused tree mortality in the red stage (e.g., Franklin et al., 2003; Coops et al., 2006; White et al., 2007), and more recently the gray stage (e.g., Dennison et al., 2010; Meddens et al., 2011). High-resolution imagery (such as QuickBird and IKONOS sensors) have the capability to detect tree mortality close to the spatial resolution of an individual tree but have often limited spatial coverage, whereas medium- and coarse-resolution imagery (such as Landsat and the Moderate Resolution Imaging Spectroradiometer (MODIS)) detect tree mortality at the stand/landscape scale and have wider spatial coverage. Currently, many forest disturbance detection methods using Landsat imagery produce categorical classifications identifying regions of undisturbed forest and insect-caused mortality, and sometimes include other classes (such as herbaceous, bare soil, water, etc., Franklin et al., 2003; Wulder et al., 2005). However, few studies to date include information on the severity of the disturbance (e.g., the amount of red stage) within a grid cell. A recent study showed that Landsat data could be used for prediction of continuous tree mortality within a pixel as opposed to a categorical classification (Meigs et al., 2011), providing a more sensitive characterization of outbreak dynamics.

Temporal dynamics from bark beetle outbreaks are often described as having four stages, including endemic, incipient-epidemic, epidemic (i.e., outbreak), and post-epidemic stages (Safranyik and Carroll, 2006). However, few studies quantify the duration of tree mortality within an area larger than a stand. Bark beetle-caused tree mortality lasted approximately 6 years within two stands (two-square mile units) in the Teton National Forest and Teton National Park in northwestern Wyoming (Cole and Amman, 1980), with bark beetles attacking progressively smaller diameter trees. Past outbreaks in western Canada have lasted on average 8–9 years with a possible minimum of 4 and a maximum of 18 years (Safranyik et al., 1974). Significantly, no study to date has assessed temporal patterns of tree mortality across larger spatial extents. Doing so will provide an improved characterization of the variability of outbreak dynamics. Yearly medium-resolution satellite observations provide an opportunity to study these temporal characteristics (e.g., Meigs et al., 2011; Walter and Platt, 2013).

Spatial patterns of bark beetle outbreaks have been described from the stand level (Mitchell and Preisler, 1991; Nelson et al., 2014) to the regional level (e.g., the entire province of British Columbia, Canada (Aukema et al., 2006; Chen, 2013)). Aerial surveys in the US and Canada document forest disturbance at regional to national scales (USDA Forest Service, 2012), and are used by managers and scientists to assess damage in forests. When used in quantitative ecological studies, these surveys are subject to uncertainty for three reasons. First, the unknown locations of damaged trees within recorded polygons makes assessing year-to-year dynamics difficult at stand scales. Second, the subjective nature of the surveys confers uncertainty (Meddens et al., 2012). Third, although flown in many regions annually, aerial surveys may not

regularly cover national parks or wildernesses. Satellite remote sensing studies offer the opportunity for accurate and repeated mapping of tree mortality, and Landsat-resolution (30-m) imagery may be useful for studying mortality dynamics within stands. However, past studies have focused on developing and evaluating methods (e.g., Meigs et al., 2011; Wulder et al., 2006b; White et al., 2005); to date, no Landsat-based studies have quantified spatial and temporal patterns of beetle-caused tree mortality at the scale of an outbreak.

The mountain pine beetle outbreak in Colorado has gained much attention recently. Warmer conditions and anomalously dry years in 2001 and 2002 have been linked to the outbreak (Chapman et al., 2012; Creeden et al., 2014). Based on aerial survey information, mountain pine beetle-caused tree mortality in Colorado peaked in 2008 with 1.15 million ha affected (USDA Forest Service, 2009) and, although newly affected areas are decreasing, tree mortality is now occurring in the lower-elevation ponderosa pine forests on the eastern side of the Rocky Mountains near densely populated areas (USDA Forest Service, 2012). In 2010, the US Forest Service allocated approximately \$30 million for mitigating impacts of the mountain pine beetle outbreak in the Rocky Mountain region.

Our objectives were (1) to develop methods to produce a continuous measure of bark-beetle caused tree mortality (a) that can be extrapolated to regions outside of the area of our reference data and over a longer period, and (b) whose predictions contain information about the severity of tree mortality within the grid cell; and (2) analyze spatial and temporal patterns of tree mortality in northcentral Colorado created by applying the continuous measure of tree mortality developed in the first objective. We developed regression models from Landsat imagery to predict percent tree mortality caused by bark beetles, taking advantage of an existing fine-resolution classification of tree mortality for model building and evaluation. We then applied the model to a Landsat scene in northcentral Colorado in 1996–2011 and quantified mortality patterns in time and space at scales from an individual 30-m grid cell to the entire Landsat scene (~10,000 ha). We report tree mortality at the year of detection rather than the year of beetle attack, which occurs 1 year earlier based on a 1-year life cycle of mountain pine beetle. This study extends our past Landsat-based research in this study area (Meddens et al., 2013) by developing a model of continuous tree mortality and by analyzing spatial and temporal patterns of tree mortality.

2. Methods

2.1. Study area

The study area is located in northcentral Colorado and in southern Wyoming (Fig. 1). The overlapping footprint of 16 years of Landsat images of Path 34/Row 32 (Worldwide Reference System 2) delineated the study area. The topography ranged from high-elevation (alpine) mountains to the plains east of the Rocky Mountains. Forests within the study area included: (a) mixed conifer forest at the higher elevations (with tree species such as Engelmann spruce (*Picea engelmannii*), subalpine fir (*Abies lasiocarpa*) and limber pine (*Pinus flexilis*)); (b) lodgepole pine (*Pinus contorta*) and occasional aspen (*Populus tremuloides*) at middle elevations; and (c) ponderosa pine (*Pinus ponderosa*) and Douglas-fir (*Pseudotsuga menziesii*) at lower elevations. The average annual precipitation is 51.6 cm (1971–2000), with average annual maximum and minimum temperatures of 9.8 and –10.2 °C (Western Regional Climate Center, Fraser station, elevation 2609 m; <http://www.wrcc.dri.edu>; accessed 6 April 2009). The study area location was chosen because of an extensive

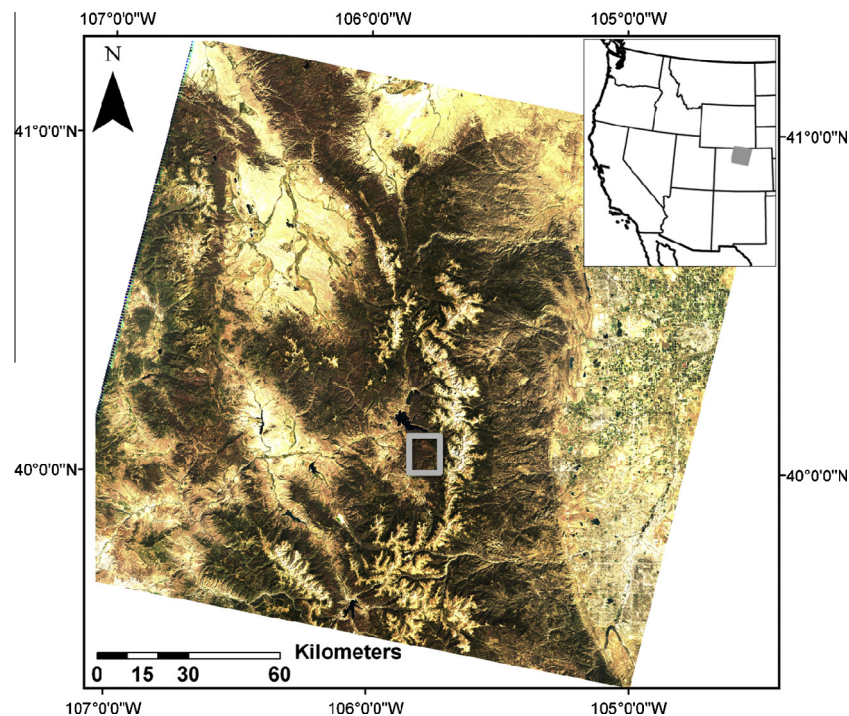


Fig. 1. Study area location in the western United States (upper right; gray polygon) and true color Landsat path 34/row 32 image from August 21, 2009. The reference area is indicated by the gray rectangle near the center of the map. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

mountain pine beetle outbreak that started around 2000 and because our previous studies provided classified, very fine-resolution aerial imagery within the area (Meddens et al., 2011).

2.2. Landsat data

We used yearly Landsat imagery data from 1996 to 2011 to predict percent red stage tree mortality within a Landsat pixel. Sixteen Landsat Thematic Mapper (TM) 5 and four Landsat Enhanced Thematic Mapper (ETM+) 7 images were downloaded from the USGS GLOVIS website (<http://glovis.usgs.gov>; accessed: 13 September 2011) and radiometrically normalized (described in Meddens et al., 2013). The images were chosen to minimize cloud cover and close to the middle of the growing season. Substantial cloud cover occurred in the imagery of 1999, 2004, 2007, and 2008, and so we used a second image in those years to fill areas of cloud cover of the first image.

We used the six shortwave Landsat bands (Band 1–Band 5, and Band 7; we did not include the thermal band (Band 6)), the tasseled cap brightness, wetness and greenness indices (Crist and Cicone, 1984; Crist, 1985), and calculated several additional spectral

indices for use as potential explanatory variables (Table 1). These indices were chosen because previous research has shown that they are effective in the characterization of land surfaces using satellite data (e.g., Tucker, 1979; Cohen et al., 1998; Wilson and Sader, 2002; Coops et al., 2006; Meddens et al., 2008; Meddens et al., 2013). In addition to the spectral indices listed above, we also calculated their anomalies in time, e.g., B5/B4' or NDVI'. The spectral anomaly (*SI'*) at a given year is the difference between the multi-temporal undisturbed mean, identified through an iterative process that removes years with disturbances, and the value of that pixel in a given year (described in Meddens et al., 2013).

2.3. Reference data

A classified aerial image of 94-km² acquired in August 2008 in the middle of the study area was used for model development and evaluation. Classes included undisturbed forest, red stage and gray stage tree mortality, herbaceous vegetation, bare soil, and shadow/water, and the classification accuracy was 90% (kappa = 0.88) (Meddens et al., 2011). We aggregated this classification to 30-m spatial resolution to match the Landsat imagery and

Table 1

Spectral bands and indices considered as potential explanatory variables in the model to predict percent tree mortality (bands (B) refer to Landsat's band order).

Index	Formula	Abbrev.	References
Red green index	$\frac{B3}{B2}$	RGI	Coops et al. (2006)
Normalized difference vegetation index	$\frac{(B4 - B3)}{(B4 + B3)}$	NDVI	Tucker (1979)
Band 5/Band 4	$\frac{B5}{B4}$	B5/B4	Vogelmann (1990), Vogelmann and Rock (1988)
Normalized difference moisture index	$\frac{(B4 - B5)}{(B4 + B5)}$	NDMI	Wilson and Sader (2002)
Tasseled cap brightness	$0.2043 * B1 + 0.4158 * B2 + 0.5524 * B3 + 0.5741 * B4 + 0.3124 * B5 + 0.2303 * B7$	TCBRI	Crist (1985)
Tasseled cap greenness	$-0.1603 * B1 - 0.2819 * B2 - 0.4934 * B3 + 0.7940 * B4 - 0.0002 * B5 - 0.1446 * B7$	TCGRE	Crist (1985)
Tasseled cap wetness	$0.0315 * B1 + 0.2021 * B2 + 0.3102 * B3 + 0.1594 * B4 - 0.6806 * B5 - 0.6109 * B7$	TCWET	Crist (1985)

produced class proportions (i.e., the percentage of each class within a 30-m pixel); we refer to this aggregated aerial image classification as reference data.

2.4. Statistical modeling of percent tree mortality

We developed and evaluated regression models for predicting the continuous percentage of bark beetle-caused tree mortality within a grid cell. Because all pixels are the same size (900 m²), predicting percent red stage tree mortality is equivalent to using area. The predicted percent mortality within a grid cell is thus equal to the canopy area of killed trees (minus overlapping crowns (Gill et al., 2000)) and closely related to killed basal area (e.g., Macomber and Woodcock, 1994). As our response variable, we used the percent red stage mortality, and developed our model with mortality from the aggregated reference classification. To focus on red stage mortality, we selected aggregated reference data for model building that had <2% gray stage and >2% red stage mortality. In addition, we used a 2008 Landsat image with multiple forest and nonforest classes (Meddens et al., 2013) to eliminate nonforest pixels. We used the reference data to select a random sample of 20 Landsat pixels within each 10% red stage mortality bin (i.e., 0–10%, 10–20%, etc.) determined by the aggregated reference data. We identified two of these sets of random samples, one for model building and one for model evaluation. This sampling method balanced the number of values across the range of mortality. The random selection was then repeated ten times to obtain ten sets of pixels for model building and ten sets for evaluation.

Our initial model preference was to use a linear model:

$$Y = \beta_i X_i + \varepsilon_i \quad (1)$$

where Y is the predicted percent red stage tree mortality within a Landsat grid cell, X_i is a Landsat-derived explanatory variable, β_i is the corresponding fitted model parameter, and ε_i is the error term. However, after we found slight yet significant spatial autocorrelation in the residuals of the linear models, we also developed a set of generalized additive models (GAM) to assess the effect of spatial autocorrelation. We included a spatial term in the GAMs that may account for spatial variability not included in the Landsat imagery while forcing all other variables to be linear:

$$Y = f(utm_x, utm_y) + \beta_i X_i + \varepsilon_i \quad (2)$$

where $f(utm_x, utm_y)$ indicates the smoothing function of the spatial term with utm_x and utm_y coordinates (Universal Transverse Mercator coordinate system) of each pixel. Although using a spatial term within a GAM is an appropriate method to analyze underlying spatial variability (e.g., Wood and Augustin, 2002; Bright et al., 2012), the inclusion of this term results in a model that should not be extrapolated outside the area in which the model was developed, thus limiting its utility.

We performed four steps to select the best candidate explanatory variables to include in our final model. First, to reduce the total number of possible explanatory variables, we considered only one SI' . We selected the single SI' that explained the most variation (i.e., the highest R^2), using one-variable (linear) models. Second, we performed a best subsets regression analysis using the linear model, which builds models for all combinations of explanatory variables. We selected the model that produced the lowest Akaike information criterion (AIC) value (a lower AIC indicates better model performance) (Akaike, 1969). Third, we only accepted a model with a variance inflation factor (VIF) < 5, similar to Jensen et al. (2006). The VIF quantifies the degree of collinearity among regression coefficients, where a VIF close to one indicates no collinearity among explanatory variables. Finally, to account for any effects of sampling for model building and evaluation, the best subsets analysis was repeated ten times with ten different random

sets, and the combination of explanatory variables that was ranked highest by the best subsets regression among sets of samples was chosen as the optimal model. After the best combination of explanatory variables was determined, we averaged the regression estimators (β_i) for each of the ten different random sets. Similarly, we averaged the regression estimators (β_i) for each of the ten different random sets using GAM for the variables that were forced linear. The statistical modeling was performed with the R statistical package (R Development Core Team, 2008).

We developed the regression models on those pixels within the region of the high-resolution imagery and then applied the (linear) model across the entire Landsat scene. We predicted percent red stage only for those pixels that contained killed trees. We used a time series of red stage pixels identified by a multi-date classification using B5/B4' for this study area from previous research (Meddens et al., 2013). From this data set, for red stage pixels, we used the model developed here to estimate the percent red stage tree mortality in red stage years. We also estimated the percent red stage tree mortality for up to 2 years before the first red stage year because we assumed that tree mortality is undetectable by the previous multi-date classification method for the first 2 years of outbreak (because of low amounts of mortality in early years). Live forest, herbaceous, and nonvegetation classes from the multi-date classification (Meddens et al., 2013) were used for locations for which we did not predict tree mortality. Other disturbances (such as fire and logging) were visually inspected and were masked out using thresholds of the tasseled cap brightness (Meddens et al., 2013). Percent red stage tree mortality predictions of <0% or >100% that erroneously occurred because of modeling errors were set to 0% and 100%, respectively. In addition, we required that percent mortality within a grid cell only increase over time.

We separately analyzed tree mortality associated with outbreak expansion into unattacked stands (i.e., grid cells with no preceding mortality) and tree mortality that intensified in already-attacked stands (i.e., grid cells with mortality in preceding years). We also estimated the number of trees killed by using the mean crown area of mature lodgepole pines killed by mountain pine beetles (9.28 m²; 95% confidence interval: 6.38–12.18 m², $n = 33$ trees), which was computed from field observations in the study area (Meddens et al., 2011).

2.5. Beetle outbreak metrics

To assess temporal patterns of the mountain pine beetle outbreak, we calculated three metrics: cumulative tree mortality (%), duration of tree mortality (years), and average rate of tree mortality (%/year) (Fig. 2). Cumulative mortality was the cumulative amount of tree mortality in 2011 calculated by the multiplication of the percent mortality and the grid cell area. The duration of tree mortality is the period from the start of tree mortality (greater than 1%) to the first year with maximum cumulative mortality. The average rate of mortality is the average annual mortality during the outbreak period divided by the duration of tree mortality. Note that during an outbreak the rate of mortality can change over time (e.g., Fig. 2), however the mean rate of mortality (averaged over the years of outbreak) can be valuable to forest managers to estimate the number of trees killed across a given spatial domain over a given time period.

These metrics change as a function of spatial resolution because more unattacked area is typically included within a grid cell with coarser spatial resolution. To quantify this change, we aggregated the 30-m resolution (grid cell area ~0.1 ha) mortality to six coarser resolutions that range across plot, stand, and landscape scales. Commonly used fixed area plots are generally between 0.001 and 1 ha (Husch et al., 2003); Gustafson and Crow (1998) and Siegert and McCullough (2003) reported average forest stand sizes of 1.2,

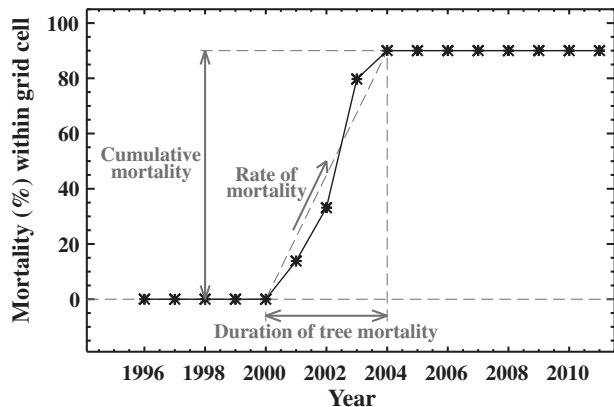


Fig. 2. Example showing the three temporal characteristics of tree mortality used in this study.

9, and 22.8 ha; and we used the coarsest resolutions to represent landscape scales. The different resolutions were thus defined as follows: 0.1 ha and 0.4 ha grid cell sizes as plot level; 1 ha, 10 ha, and 1 km² as stand level; and 10 km² and 100 km² as landscape level. The three temporal metrics were recalculated for each resolution using aggregated grid cells that experienced at least 1% cumulative tree mortality (thus, coarser resolutions included unattacked areas that were excluded at finer resolutions). We focused on grid cells in which outbreaks had ended (i.e., minimal beetle activity at the end of the study period), and so we excluded grid cells in which tree mortality increased from 2010 to 2011 by >2%. Because forest composition and structure varies with elevation, we also assessed mortality patterns across different elevation ranges. We downloaded the 30-m USGS digital elevation model (DEM) (<http://data.geocomm.com/dem>, accessed: November 22, 2010) for the study area.

3. Results

3.1. Prediction of percent mortality

B5/B4' was the spectral anomaly that best predicted percent mortality at the 30-m pixel resolution and was included as the only spectral anomaly candidate in the final model. The combination of B1, NDVI, and B5/B4' was chosen most often using the ten-times repeated best subsets regression. For the linear regression, the explanatory variables B1, NDVI, and B5/B4' were highly significant ($p < 0.001$). For the GAM, NDVI, B5/B4', and the spatial term were highly significant ($p < 0.001$) and the B1 was slightly less significant ($p < 0.01$). Means of fitted parameters ($\beta_1 - \beta_n$) of the GAM changed <18% compared with those of the linear model; the exception was the fitted parameter for B1, which changed more than 60% (Table 2). As described above, we found a slight degree of spatial autocorrelation within the residuals of the linear model (average global Moran's I: 0.07, $p < 0.01$ with distance matrix: $w_{ij} = 1/d_{ij}$,

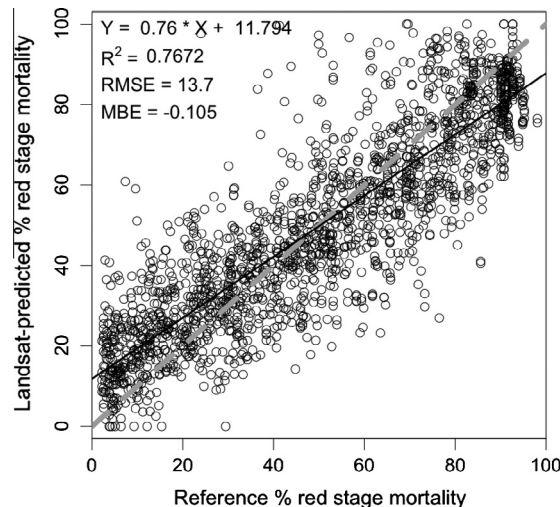


Fig. 3. Comparison between reference (aerial imagery) and Landsat-predicted percent red stage tree mortality within 30-m grid cells using multiple linear regression models. Solid line is the regression line; gray dashed line is the 1:1 line; RMSE, root mean square error; MBE: mean bias error.

whereby w_{ij} represents the weighting matrix with d_{ij} the distance between pixel i and j). The GAMs, however, showed no significant spatial autocorrelation in the residuals (average global Moran's I: 0.02, $p > 0.05$).

Reference (aerial image-derived) versus model predictions (from evaluation samples) of percent red stage tree mortality resulted in an overall mean R^2 of 0.77 for the linear model (Fig. 3) and 0.80 from the GAM (Appendix A). The predicted values between the linear model and the GAM showed strong agreement ($R^2 = 0.94$; Appendix B), and inclusion of a spatial term increased the explained variance by 3.3% based on the R^2 . However, extrapolation of GAM percent red stage tree mortality predictions is not valid for locations outside of the model development area. Therefore, to predict red stage tree mortality for the entire Landsat footprint, we used the linear model.

3.2. Tree mortality patterns

We estimated percent red stage tree mortality for each year in the Landsat time series (1996–2011) using the linear model (Table 2) as described in Section 2.4. Severe beetle-caused tree mortality started in the region around 2003 at the center of the Landsat scene (Fig. 4). From this location, beetle-caused tree mortality spread north and east. Large areas of tree mortality occurred in 2007–2009 (Fig. 5a). Estimated tree mortality in 2008 was lower than the preceding and proceeding years, likely due to increased cloud cover that obscured tree mortality. Cumulative tree mortality area in 2011 was 211,709 ha, or 22.1% of the forested area within the study area (Fig. 5b). Earlier in the outbreak, the

Table 2
Summary of the two regression models that predict percent red stage tree mortality within 30-m pixels. Values represent averages over ten-times repeated model building and evaluation ($n = 200$ per building or evaluation set). R^2 and root mean square error were determined from observed versus predicted values.

Model	R^2	RMSE ^a	Moran's I ^b
Linear model (% Red attack/100) = 1.66 – 11.00·B1 – 1.87·NDVI + 0.66·B5/B4'	0.77	13.7	0.073
GAM model (% Red attack/100) = 1.39 + $f(utm_x, utm_y)$ + 4.66·B1 – 1.59·NDVI + 0.59·B5/B4'	0.80	12.7	0.017

^a Root mean square error.
^b Global Moran's I of the residuals.

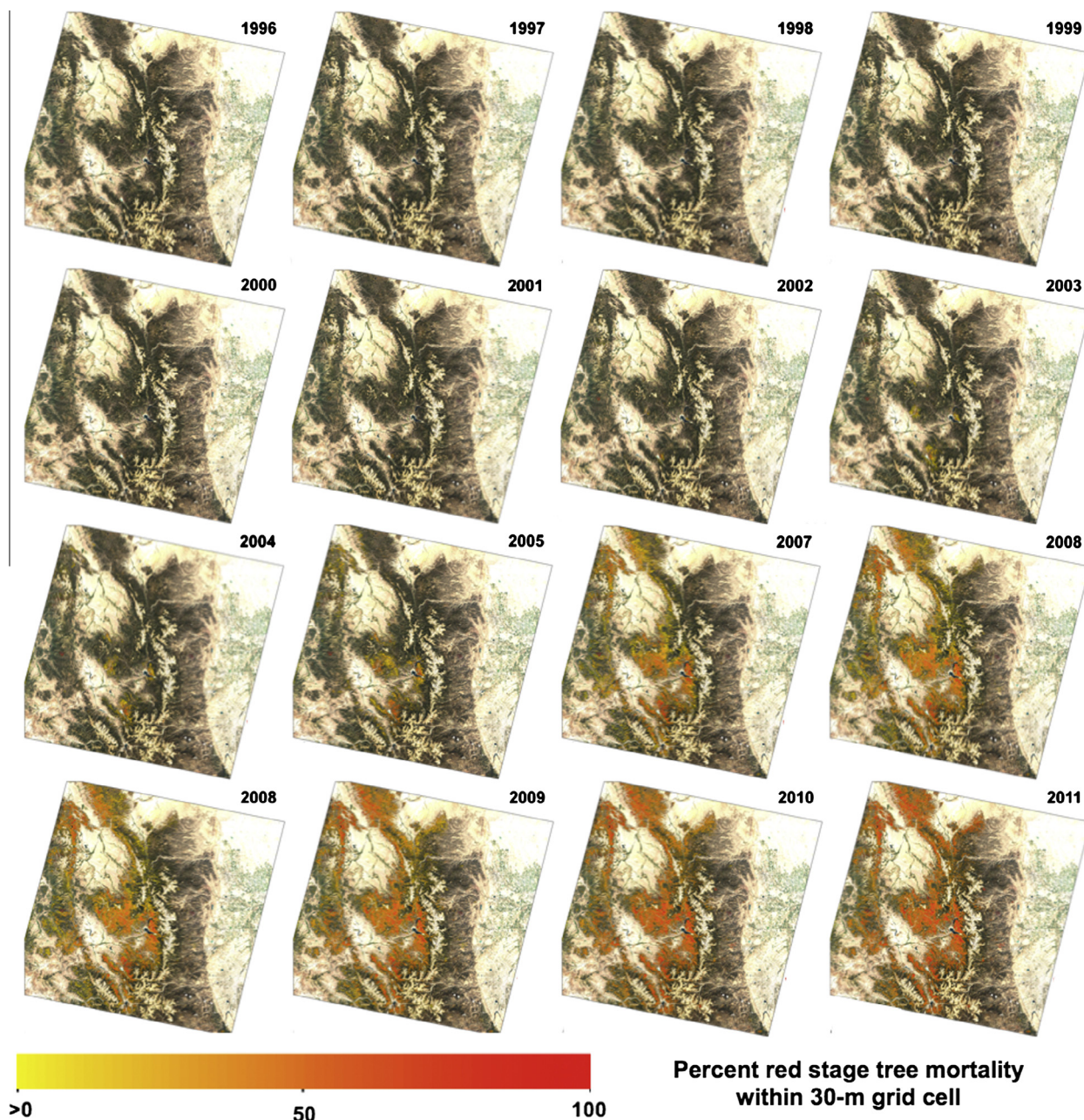


Fig. 4. Percent red stage tree mortality (within 30-m grid cell) for the Landsat time series (1996–2011) and a true-color Landsat background image from 2002. Note that year of detection of tree mortality is shown, which is approximately 1 year after beetle attack. See [Appendix C](#) for an animation of the time series of detected tree mortality. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

majority of the tree mortality occurred as a result of expansion into unattacked stands, whereas later in the outbreak the majority of the tree mortality occurred as intensification of mortality in already-attacked stands ([Fig. 5a](#)). For the last years of our time series (2010–2011), tree mortality was declining, and almost all new mortality was associated with intensification of existing outbreaks, not expansion. The total estimated number of trees killed within this outbreak period and study area was 228 million mature lodgepole pine trees with a range of 174–332 million trees using the 95% confidence interval of 33 field-measured crown areas within the study area ([Table 3](#)).

The intermediate elevation range (2500–3000 m), the elevation of predominantly lodgepole pine in northern Colorado ([Stohlgren and Bachand, 1997](#)), had the highest average mortality percentage

(29.4%) ([Table 3](#)). The lowest (mainly dominated by plains and open ponderosa pine trees) and highest (dominated by subalpine fir, Engelmann spruce, and limber pine) elevations had low percent mortality (2.7% and 4.2%, respectively; [Table 3](#)). Elevations of 2000–2500 m that consisted of ponderosa pine, Douglas-fir, aspen, and to a lesser extent lodgepole pine had 5.2% mortality, and 3000–3500 m, dominated by spruce-fir forests with lodgepole pine intermixed (especially in the lower part of the range), had 16% mortality.

Within the reference data area, locations with lower tree mortality in 2005 (<40%) and more severe tree mortality in 2006 (40–80%) match the locations of gray stage trees identified in the 2008 classified aerial image ([Meddens et al., 2011](#)), indicating that the continuous measure of bark beetle-caused tree mortality not

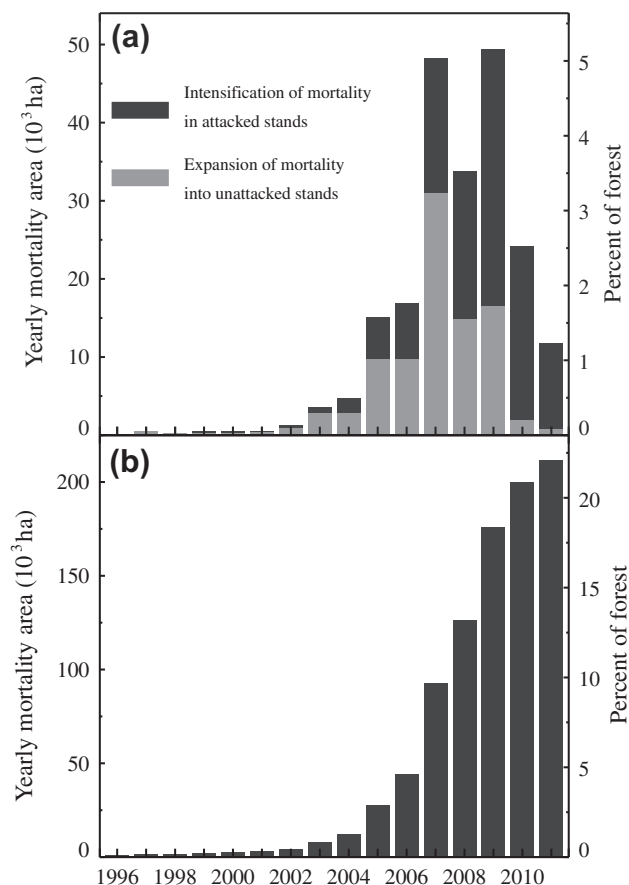


Fig. 5. (a) Yearly and (b) cumulative tree mortality area (10^3 ha) (calculated by summing the product of the percent mortality and 30-m grid cell area) caused by bark beetles for the Landsat time series (left axes) and as a percentage of the total forest within the scene (right axes). Note that the 2008 imagery contained substantial cloud cover. Year of detection of tree mortality is reported, which is 1 year after beetle attack. In (a), areas were divided by expansion (grid cells with no prior tree mortality) versus intensification (with prior tree mortality).

only captured the severity but also the timing of the mortality (assuming a transition of 2–3 years from red to gray stage; Saf-ranyik, 2004; Wulder et al., 2006a).

3.3. Temporal and spatial metrics of beetle outbreak

Grid cells with the 30-m spatial resolution within the outbreak experienced on average 60% cumulative tree mortality (Fig. 6a). The average duration of the outbreak at the 30-m spatial resolution was 3–4 years (Fig. 6b) and the average rate of mortality was 20%/year (Fig. 6c).

Metrics describing temporal characteristics of the outbreak varied substantially by grid cell size. Cumulative tree mortality for attacked grid cells decreased when increasing the size of the

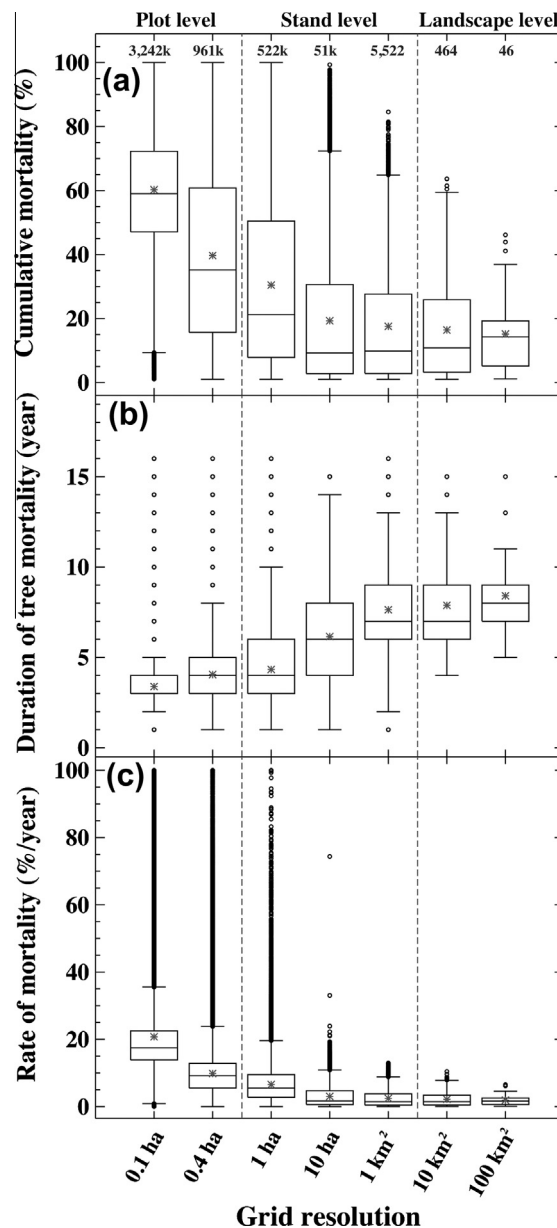


Fig. 6. (a) Average cumulative tree mortality (%), (b) duration of tree mortality (year), and (c) average rate of mortality (%/year) within grid cells with mortality >1% across different grid resolutions. Star symbols indicate the mean and numbers across the top indicate the number of grid cells (n ; with $k = \times 10^3$).

spatial resolution (i.e., from plot-level to landscape-level) (Fig. 6a). The decrease largely occurred because area with non-host or non-susceptible trees (e.g., younger, smaller diameter lodgepole pine trees) was added in coarser-resolution grid cells; these areas

Table 3

Cumulative beetle-caused tree mortality (calculated by summing the product of the percent mortality and 30-m grid cell area) in 2011 across different elevations.

Elevation range	Area of tree mortality (1000 ha)	Percentage of forest (%)	Number of killed trees ^a (millions)
<2000 m	0.33	2.7	0.35 (0.3–0.5)
2000–2500 m	5.65	5.2	6.1 (4.6–8.9)
2500–3000 m	158.2	29.4	170.5 (123.0–247.9)
3000–3500 m	47.3	16.0	50.9 (38.8–74.0)
>3500	0.23	4.2	0.3 (0.3–0.4)
All elevations	211.7	22.1	228.2 (173.9–331.7)

^a Area of tree mortality divided by average (and $\pm 95\%$ confidence interval) of the tree crown area of mature lodgepole pine trees (9.28 m^2 ; 95% confidence interval: $6.38\text{--}12.18 \text{ m}^2$, $n = 33$ trees), computed from field observations in the study area (Meddens et al., 2011).

were excluded from the analysis of the finer-resolution grid cells. The variability of the cumulative mortality across the study area was large regardless of grid cell resolution (Fig. 6a).

The duration of tree mortality within attacked grid cells decreased when increasing the spatial resolution (Fig. 6b). This indicates that for the plot level, the outbreak lasted 4 years in individual stands, whereas for larger areas (such as a national forest) the outbreak lasted 8 years. There was considerable variability that indicated that some areas experienced a short duration of mortality, whereas other areas experienced slower mortality. By analyzing the spatial patterns of the duration of tree mortality (not shown), we found that in the early years (2002–2004) the duration of outbreak within stands was longer, whereas when beetle populations were high and host availability was abundant (2005–2008), the outbreak was shorter within a stand.

The average annual rate of tree mortality for beetle-attacked locations decreased when increasing the spatial resolution (Fig. 6c). At finer spatial resolutions, mortality occurs more rapidly than across larger areas, related to host depletion, beetle pressure and dispersal, and tree species composition. In addition, the variability at smaller spatial scales was much larger possibly related to greater difference between forest cover types between the different grid cells.

4. Discussion

We developed an accurate continuous measure of bark beetle-caused tree mortality using Landsat spectral imagery and finer-resolution reference data. Continuous measures of tree mortality provide more information than a categorical classification (undisturbed versus disturbed classes), thereby increasing the usefulness of these data for some ecological analyses. Using a linear model, we showed good agreement between predictions derived from Landsat spectral bands and observed percent red stage tree mortality derived from reference data.

Bark beetle-caused tree mortality was highest in 2006–2009. By the end of the study period (2011), cumulative tree mortality covered a substantial amount of forest (22%) across the study area. The initial detection year of severe mountain pine beetle-caused tree mortality was around 2003, indicating that the start of the major beetle attack occurred in 2002 (note that beetles attack 1 year before needles turn red), co-occurring with warming conditions and a multi-year drought (2000–2002) in the study area (Chapman et al., 2012; Creeden et al., 2014) that reduced tree defenses (Raffa et al., 2008). We found rapid expansion of tree mortality into new locations in the earlier stages of the outbreak and increased intensification of already-attacked locations later in the outbreak, possibly linked to an initial large number of susceptible hosts trees followed by host depletion (that reduced expansion opportunities). During an outbreak, large-diameter trees are attacked first, but when host availability decreases smaller trees are attacked in subsequent years until eventually the beetle population collapses (Safranyik and Carroll, 2006).

A range of cumulative mortality occurred, although even in this major outbreak, tree mortality rarely reached 100%, indicating that there are still living trees within the forest confirming field studies within parts of our study area (Klutsch et al., 2009; Nelson et al., 2014). Our average 60% plot-level cumulative tree mortality (by canopy area) was similar to other published amounts within the study area. Field observations of mean basal area of killed trees were 56% and 64% of preoutbreak total basal area (Klutsch et al., 2009 and Nelson et al., 2014, respectively). Mountain pine beetle-caused tree mortality at the finest scale was 3–4 years, faster than the reported 6 years by Cole and Amman (1980), although our rates of mortality were similar to those from field observations

(Nelson et al., 2014). Differences may be related to difference in spatial extent over which the duration of mortality is determined and the high beetle pressure in our study area. Our estimated outbreak duration is relatively rapid compared with other insects such as defoliators that cause tree damage for periods often longer than 6 years (Meigs et al., 2011), but relatively slow compared to more immediate forest disturbances such as fire and harvest. The average duration of tree mortality at the coarsest scale (8.4 years) approached the average duration of an outbreak (8–9 years) reported by Safranyik et al. (1974). These temporal spectral characteristics can be used to improve mapping at the Landsat resolution. Future research on the detection of beetle-caused tree mortality at coarser spatial scales (e.g., MODIS data) should consider that tree mortality at this resolution progresses slower within a single grid cell than at finer resolutions, possibly leading to delayed detection and leading to lower overall cumulative percent mortality at the end of an outbreak.

The linear model accurately captured mountain pine beetle-caused tree mortality. However, this model was not able to account for all spatial variability within the reference area, and a generalized additive model with a spatial component slightly increased prediction accuracy. Although we preferred the GAM for locations inside the reference area, the linear model was more appropriate for extrapolation outside the reference area as described in the Results. The residuals in our linear model showed spatial autocorrelation, possibly impeding calculated statistics from our model (Lennon, 2000) and reducing the accuracy of predictions from the linear model. Nonetheless, we found that predictions of the GAM resulted in very similar predictions as our linear model, warranting the use of the linear model.

We found some tree mortality that was not correctly attributed to the mountain pine beetle. For instance, a small amount of detected tree mortality in the northeastern part and the higher elevations the study area corresponded with areas identified by USDA Forest Service aerial detection surveys as spruce beetle-caused tree mortality. Subsequent analysis of Landsat data should therefore focus on automatically attributing different mortality agents within a single scene. Spectral responses of lodgepole pine and ponderosa pine killed by mountain pine beetles appear slightly different (Ciesla, 2006), and the spectral response of spruce trees killed by spruce beetle is distinctive from large-scale mountain pine beetle disturbances. Finally, because the red stage often lasts longer than 1 year, studies that focus on the ability to detect different stages of tree mortality (such as red versus gray stage tree mortality) will improve the understanding of the timing of tree mortality.

5. Conclusions

In this paper, we used multi-temporal satellite data to analyze patterns of mountain pine beetle-caused tree mortality. With annual time series of detected tree mortality we could more completely describe this bark beetle outbreak than using single-date imagery. Grid cells of 0.09 ha exhibited large ranges of mortality severities, which are generally not captured in a binary classification. Using regression models we predicted percent tree mortality within a grid cell, which led to a more resolved representation of tree mortality than a binary classification approach. The outbreak in Colorado began around 2000, peaked in 2007–2009, and was declining in the last years of the study period. Beetles killed 22% of the forest within the study area.

We found that bark beetle outbreaks within 30-m grid cells lasted 3–4 years, grid cells experienced an average of 60% mortality in outbreak areas, and beetles killed an average of 20% of a grid cell per year in outbreak areas. These metrics of outbreaks decreased at larger scales in which more unattacked areas were included as grid

cell size increased. This spatial explicit information about severity and duration of a severe bark beetle outbreak across different spatial scales is useful for forest managers overseeing bark beetle-affected landscapes and can be used for comparison between other bark beetle outbreak areas. The large extent of tree mortality and high percentage of tree mortality within grid cells may be unusual, although additional studies to quantify other outbreaks of bark beetles, both within lodgepole as well as in other areas that experienced extensive tree mortality (Meddens et al., 2012), are needed. Recent warming coupled with a drought in the early 2000s may be responsible for this and other bark beetle outbreaks with similar amounts of tree mortality in recent years (Chapman et al., 2012; Creeden et al., 2014), suggesting that severe, extensive tree mortality may become more frequent in the coming decades in response to climate change if host trees remain available.

In addition to increasing the understanding of bark beetle ecology and dynamics, the methods presented here can be used in monitoring and mapping other tree mortality events. With the potential of increasing global tree mortality due to climate change (Allen et al., 2010), detection and monitoring of tree mortality will be facilitated by the use of Landsat satellite imagery. Furthermore, our Landsat-based methods can be used to aid development of algorithms for monitoring tree mortality using coarser-resolution satellite imagery (e.g., MODIS) that covers greater extent with higher temporal resolutions.

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Appendix A

see Fig. A1.

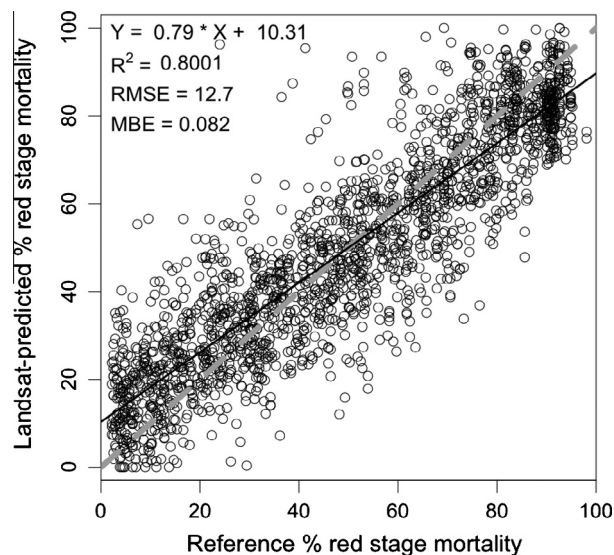


Fig. A1. Comparison between reference (aerial imagery) and Landsat-predicted percent red stage tree mortality within 30-m grid cells using a generalized additive model (GAM) that included a nonlinear term to account for spatial variation. Solid line is the regression line; gray dashed line is the 1:1 line; RMSE, root mean square error; MBE: mean bias error.

Appendix B

see Fig. B1.

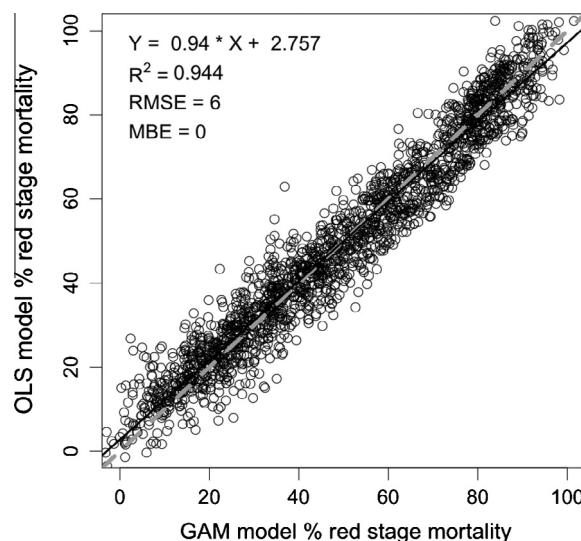


Fig. B1. Comparison between percent red stage tree mortality predictions of the multiple linear regression models and the (linear) generalized additive models (GAM) with nonlinear spatial term. Solid line is the regression line; gray dashed line is the 1:1 line; RMSE, root mean square error; MBE: mean bias error.

Appendix C

see Fig. C1.

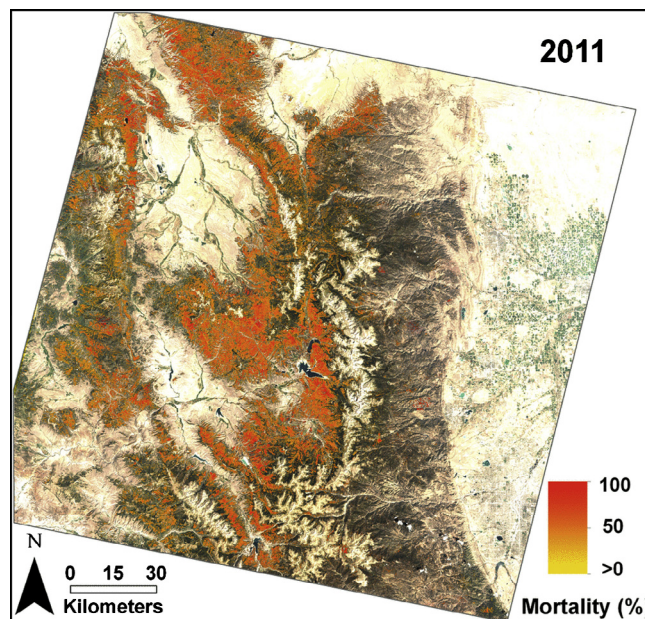


Fig. C1. Animation of the detected tree mortality (% tree mortality within 30-m grid cell) within the study area from 2000 to 2011 and a true-color Landsat background image from 2002. The year of detected tree mortality is 1 year after the beetle attack. Note: open associated .gif file with an Internet browser (e.g., Firefox) (supplementary video 1).

Appendix D. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.foreco.2014.02.037>.

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